

# Advaya Verma

+91 9319200479 | [advaya2208@gmail.com](mailto:advaya2208@gmail.com) | [LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [Portfolio](#)

## PROFESSIONAL SUMMARY

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Final-year Computer Science undergraduate at VIT-AP with hands-on experience in machine learning, NLP, and backend development. Proficient in Python, REST APIs, and cloud-native tools including Docker and FastAPI. Oracle-certified Generative AI professional with a strong foundation in data structures, algorithms, and software engineering principles. Seeking a software engineering or ML internship to apply skills in real-world AI/backend projects.

## EDUCATION

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**Vellore Institute of Technology, Andhra Pradesh** — B. Tech, Computer Science and Engineering | 2023 – 2027

**CGPA: 8.1**

**Relevant Coursework:** Data Structures & Algorithms, Operating Systems, DBMS, OOP, Machine Learning, Artificial Intelligence, Software Engineering & UML, Theory of Computation, Compiler Design, Design & Analysis of Algorithms

**Ryan International School, Faridabad** — Senior Secondary (Class XII) — CBSE | 2023 Secondary (Class X) — CBSE | 2021

## TECHNICAL SKILLS

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**Languages:** Python, C, C++, Java, R, JavaScript

**Web & Backend:** HTML, CSS, React, FastAPI, REST APIs

**ML / AI Libraries:** Scikit-learn, NumPy, Pandas, Sentence Transformers, FAISS

**Tools & Platforms:** Docker, Git, GitHub, MySQL, Oracle Cloud Infrastructure

**Concepts:** NLP, Semantic Search, Clustering, Agile/Scrum, Software Engineering, UML

## PROJECTS

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**Semantic Search System** | Python, FastAPI, FAISS, Docker, NLP, Scikit-learn | [GitHub](#)

- Built a semantic search pipeline using Sentence Transformers (MiniLM-L6-v2) and FAISS, achieving sub-100ms nearest-neighbor retrieval over 18,000+ documents from the 20 Newsgroups dataset.
- Implemented Gaussian Mixture Model (GMM)-based fuzzy clustering to assign probabilistic multi-topic memberships to documents, improving retrieval relevance for ambiguous queries.
- Designed a semantic cache layer to detect similar queries and reuse results, reducing redundant API calls and cutting average response time by ~40%.
- Exposed REST API endpoints via FastAPI and containerized the full application using Docker, enabling one-command portable deployment.

**Digital Library Web App** | HTML, CSS, JavaScript | [Live Demo](#)

- Developed a fully client-side digital library system with JSON-based local storage, supporting CRUD operations (add, search, update, delete) for book records.
- Built a responsive UI with real-time search and filtering, enabling efficient book management without a backend dependency.
- Deployed as a GitHub Pages static site, demonstrating end-to-end ownership from development to deployment.

## CERTIFICATIONS

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**Hashgraph Developer Course** — The Hashgraph Association | Mar 2026

**Oracle Cloud Infrastructure 2025 Certified Generative AI Professional** — Oracle University | Oct 2025